Specialist Interview Report
Squigglr

Team #17, Squigglr
Rayan Nasrallah
Ahmed Elzaria
Muhammad Huzaifah
Shamil Canbolat
Moustafa Moustafa

Table 1: Revision History

Date	**Developer(s)**	**Change**
11 November 2026	Rayan Nasrallah	First version: methodology, questions, subjects, empathy map analysis, results and limitations, with the transcript as an appendix (#262).

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1 Purpose

This report is the fall custom document for Squigglr. It describes the interview we held with three domain specialists on October 1, 2026, and analyses what they told us with an empathy map. The Software Requirements Specification (SRS) already traces each interview statement to the requirements it shaped (SRS Appendix A, SRC-005). This report does not repeat that trace. Instead, it looks at the interview as a whole: what the specialists said, thought, did and felt, which themes run across their answers, and what we still need to ask them. Those open questions feed the next session, where the same specialists will try the proof of concept.

2 Methodology

2.1 Interview type

We ran one semi-structured group interview. A semi-structured interview gives a fixed core of questions, so every topic we needed gets raised, while follow-up probes let the specialists take the discussion where their experience points. A structured interview would have lost the reasoning behind each answer, and an unstructured one would have been hard to keep on track with three people in 43 minutes.

2.2 Procedure

- **Format:** a remote video call of 43 minutes, recorded and transcribed automatically.
- **Interviewers:** Rayan Nasrallah (facilitator) and Moustafa Moustafa.
- **Structure:** a 10-minute presentation of the planned application (guided lessons, creative drawing space, practice history for adults, and two possible future features), then the questions in Section 4.
- **Consent:** all three specialists agreed verbally at the end of the session to the use of the transcript for eliciting requirements (41:51–42:48), and signed consent forms sent to them by email after the session.
- **Follow-up:** the questions we ran out of time for went into a short written questionnaire, answered by two of the three specialists on October 2 and 4, 2026 (SRS Appendix, SRC-009).

2.3 Data preparation

The automatic transcript was de-identified at the specialists' request: names, institutions, locations and a personal health detail were removed or replaced with bracketed placeholders. Mis-heard words were corrected and filler words removed. Each turn keeps its timestamp from the recording, so every claim in this report can be checked against the transcript in Appendix B.

2.4 Analysis method: empathy map

We analysed the transcript with an empathy map, a design thinking tool that sorts what is known about a group of people into four quadrants [1, 2]:

- **Says:** what they stated directly.
- **Thinks:** the beliefs and priorities behind what they said.
- **Does:** what they reported doing in their own work.
- **Feels:** the reactions and concerns they expressed.

We chose an empathy map for two reasons. First, the SRS already links single statements to requirements, so a second statement-by-statement coding would add nothing. An empathy map instead shows the specialists' point of view as a whole, which is what we need to plan the next session with them. Second, it separates what was said from what was reported as practice, which matters because the specialists' experience comes from their own projects, not from observing our users.

The steps were:

1. Read the full transcript and mark every turn by a specialist that carries content (not greetings or audio checks).
2. Place each marked turn in one or more quadrants, keeping its timestamp. "Says" holds paraphrases of direct statements. "Thinks" and "Feels" only hold beliefs and reactions the specialists put into words; we did not guess at unspoken ones.
3. Group the entries into themes, then lay the map out with one row per theme, so each statement sits beside the belief, practice and reaction behind it.
4. For each theme, note where the specialists disagreed, and what we still need to ask.

We built one map for the group rather than one per specialist; the empathy map section explains why.

3 Subjects

Three domain specialists took part. They research and develop software for children with autism, ADHD and other learning needs, and hold doctorates in human-computer interaction and related engineering fields. They work at the same research centre outside Canada. They preferred not to be named, so this report refers to them as P1, P2 and P3 and gives no further detail that could identify them. With only three people from one centre, we report their background as a group rather than per person.

Code	Role in the session
P1	Specialist. Asked the first questions after the presentation and led the answers on progression, child independence and input.
P2	Specialist. Led the answers on population, accessibility guidelines and audio feedback.
P3	Specialist. Raised the question of pen or finger input.

Table 2: Subjects

4 Questions

4.1 Questions asked in the session

We had prepared a guide of 16 main questions. After the presentation, the specialists raised their own questions first, and the discussion followed them. Table 3 lists the questions as they were actually asked, by us or by the specialists, in order.

Time	Asked by	Question
12:07	P1	Did you co-design the app with educators, therapists or other end users?
13:42	P1	(Comment) Camera finger tracing may target a different skill.
14:55	Moustafa	Is camera finger tracing helpful if we do implement it?
15:48	Rayan	With limited time, should we focus on shapes and tracing before the camera feature?
16:45	P2	Are you targeting specific groups of children, or a broader population?
20:14	Moustafa	So the app should not be limited to children with autism or ADHD?

Time	Asked by	Question
22:07	Rayan	Would you take the app further in your own research, or help observe children using it?
24:34	P3	Will children draw with a pen or with their finger?
27:31	Rayan	Should the app decide when a child is ready to move to the next level, or leave that to the supervising adult? What determines readiness?
31:10	Rayan	What progress information would help a parent or caregiver judge whether the child is ready to move on?
32:12	Rayan	Can you connect us with occupational therapists or educational specialists?
33:34	Moustafa	Should a child be able to use the app alone, or should we assume an adult is present? How should the app give feedback and instructions?
37:18	Rayan	Is spoken audio feedback a good idea?
38:58	Rayan	Which design choices cause problems for autistic children or children with ADHD, and which practices should we follow?
40:44	Rayan	Any other comments or patterns we should know about?

Table 3: Questions asked in the session

4.2 Coverage of the prepared guide

Table 4 shows which topics of the prepared guide were covered in the session, which went to the follow-up questionnaire, and which were not covered.

Guide topic	Covered by
First reactions	Session
Whether and how children practise these skills now	Not covered
Existing apps and tools	Not covered
What adults find hardest when guiding a child	Not covered
School tool, home tool or both	Partly (P2, 35:36)
Adult present or child alone	Session
Devices and input methods	Session (pen or finger)
What motivates or frustrates children	Questionnaire
How to give feedback	Session (audio) and questionnaire
How adults decide a child is ready for less support	Session
Design choices that cause problems	Session
Whether cleaning up free drawings helps	Questionnaire
Useful practice information for adults	Session
Camera finger tracing	Session
Letter tracing	Questionnaire
Top priorities	Not covered

Table 4: Coverage of the prepared interview guide

5 Raw Data

The de-identified transcript is in Appendix B.

6 Analysis

6.1 Empathy map

Table 5 is the empathy map. Empathy maps are often built for one user at a time and then combined into one map for a user type [2]. Instead, we built a single map for all three specialists, for two reasons. First, they answered as a group: they built on each other's answers, so splitting the session by person would separate points that were made together. Second, the specialists did not speak equally. P3 spoke on one topic (pen or finger input), so a map for P3 alone would have about two entries. Keeping one map, with every entry labelled by speaker, shows who said what without three thin maps. Section 6.2 shows who led each theme.

A classic empathy map has four quadrants. Here each row is one theme, so a statement sits beside the belief behind it, what the specialists do in their own work, and how they felt about it. Each entry is a short paraphrase with the speaker and a link to the transcript. A dash marks a quadrant the interview gave no evidence for; those gaps are discussed in Section 7.

Says	Thinks	Does	Feels
Theme 1: Adults decide, the app supports			
An adult who knows the child should choose the level (P1, 28:37). Let the teacher or caregiver change the activity and see progress (P2, 29:18).	Having the app decide means assessment, and a progress view raises the question of what assessment it applies (P1, 28:37; P2, 29:18).	Built a vocabulary app where the teacher personalized each child's path through a separate adult view (P2, 29:18).	Cautious: automatic progression is "more advanced" and too complex for now (P1, 28:37).
Theme 2: The child should not need an adult in the room			
A child who cannot read should be able to enter a lesson, practise, restart and move around using images and icons (P1, 34:13).	Children are used to icon-based games; adults handle setup (P1, 34:13). The child-and-caregiver workflow is the team's choice (P2, 35:36).	–	Concern that parents get annoyed if they must keep stepping in (P1, 34:13).
Theme 3: Design the details with the people who know the children			
Which progress measures help should be decided with occupational therapists or educators (P2, 31:42). Use spoken audio that can be turned off (P2, 37:39).	Educators and therapists, not designers, know the children day to day (P1, 36:26). Offering more than one modality is good inclusive design (P2, 37:39).	Design praise wording and colours from the literature and therapists' feedback, bit by bit (P1, 36:26). Gave their app voice input that users can turn off (P2, 37:39).	–
Theme 4: Accessible and broad from the start			
Start with WCAG, then autism design heuristics (P2, 39:26). Don't limit the app to one diagnosis (P1, 20:27).	Designing for disability serves everyone; the skills apply to all children, at different stages (P2, 19:05; P2, 20:47).	Teaches accessible design using WCAG (P2, 39:26).	P2 related to the problem from their own difficulty drawing lines as a child (P2, 16:45).

Says	Thinks	Does	Feels
Theme 5: The input tool shapes the skill			
A finger is fine if the goal is the drawing rather than fine motor skills (P3, 25:36). A finger is easiest, but fine motor practice needs a held tool, from large to small (P1, 26:46).	How a child holds a pen affects their drawing and the app's usability (P3, 25:36).	–	–
Theme 6: Core first, with care around children			
Focus on touchscreen tracing and add camera tracing later (P1, 16:13; P2, 16:45). Testing with children needs ethics approval; meeting experts does not (P2, 22:56; P1, 33:00).	Camera tracing trains gross rather than fine motor skills, but can help a child visualize the shape (P1, 13:42; P1, 15:12).	Obtains ethics approval for studies with children, which takes about three to four months (P2, 22:56).	Willing to keep helping: to meet again, guide a pilot and share contacts (P1, 24:03; P1, 41:08; P2, 22:56).
Overall			
–	–	–	Positive about the project: "very nice", "inspiring", "much needed", "timely" (P2, 16:45; P1, 24:03; P1, 41:08).

Table 5: Empathy map of the three specialists, by theme

6.2 Themes

1. **Adults decide, the app supports.** Every quadrant agrees: the adult chooses the level, because anything else becomes an assessment.
2. **The child should not need an adult in the room.** This pulls against theme 1: the adult controls progression but must not have to sit through every session.
3. **Design the details with the people who know the children.** Feedback wording, colours, audio and progress measures should come from occupational therapists and educators.
4. **Accessible and broad from the start.** WCAG first, then autism heuristics, for all children rather than one diagnosis.
5. **The input tool shapes the skill.** The choice depends on the goal: P3 favoured a finger if the goal is the drawing itself, and P1 agreed a finger is the easiest start but said fine motor practice needs a held tool.
6. **Core first, with care around children.** Touchscreen tracing before camera tracing, and expert feedback rather than testing with children.

Table 6 shows which specialist led each theme.

6.3 How the themes were used

We built what the specialists told us, as our supervisor advised. Themes 1, 2, 4 and 6 became requirements and constraints in the SRS. For theme 5 we followed P1: the stylus is the main input because fine motor

Theme	Who led it
1. Adults decide	P1 first, then P2 from a past project
2. Child without an adult	P1; P2 added that the workflow is the team's choice
3. Design with those who know the children	P1 (feedback wording and colours) and P2 (audio, progress measures)
4. Accessible and broad	P2; P1 agreed
5. Input tool	P3 raised it; P1 answered
6. Core first, care around children	P1 (camera tracing, ethics); P2 agreed and gave the ethics timeline

Table 6: Who led each theme

practice is the goal, and a finger still works. Theme 3 is only partly met, and here we consulted our supervisor. The specialists suggested deciding progress measures with occupational therapists (P2, 31:42). Our supervisor advised against waiting on one for the values that depend on the child, such as how accurate a drawing must be to pass a stage and how many failed attempts end a lesson, because these differ from child to child, so an occupational therapist could not give one right value either (supervisor meeting, October 14, 2026). Instead, the adult who knows the child sets these values for that child, within a minimum and maximum the team defines and shows with example drawings, and the midpoint is the default. This avoids the risk of the team fixing values that suit some children but not others, which we are not qualified to judge. It also follows theme 1, since the adult already decides the child's level. In the same way, the adult chooses which recorded values their progress view shows. Feedback wording and colours will be reviewed with the specialists in the follow-up session. The SRS records the trace from each statement to its requirements (SRS Appendix A, SRC-005.01 to SRC-005.13).

7 Results and Next Steps

The map shows that the specialists' view is consistent: a child-friendly app that the child can use alone, controlled by the adult who knows the child, accessible from the start, and designed with the people who work with children every day. Their main reservation is any feature that looks like an assessment.

The dashes in the map show what we did not learn. The "Does" column is empty for child independence and input, because the guide's questions on how children practise now and which tools exist were not reached. The "Feels" column says little about feedback and input, and nothing about how children react, because the specialists spoke from their own projects.

We will return to the same specialists with the proof of concept, a working lesson they can try on their own time. That session will check our reading of each theme, and ask about the gaps:

- Can a child who cannot read get through the lesson with the icons and audio we chose (themes 1 and 2)?
- Are the feedback wording, colours and sounds suitable (theme 3)?
- With the stylus and a finger both available, does the stylus still seem the right main input (theme 5)?
- Does the adult's progress view read as an assessment (themes 1 and 3)?
- How do children practise these skills now, and which existing tools fall short (not covered in the first session)?

In that session we will run the demo first and explain our decisions only after it ends, so the specialists react to the app rather than to our reasoning.

8 Limitations

- **Specialists only.** We interviewed three specialists. We did not interview parents, teachers or occupational therapists directly, and we did not involve children.
- **Expert opinion, not observation.** The specialists described their own experience and projects. Nothing here comes from observing children use Squigglr.
- **One session, one centre.** All three work at the same centre, in one 43-minute session, so their views may share a common background.
- **Group and presentation effects.** In a group, one answer can shape the next, and presenting our design first may have led the specialists to react to our features rather than raise needs we missed.
- **Inferred quadrants.** "Thinks" and "Feels" rely on what the specialists put into words. We did not infer unspoken feelings, so these quadrants are incomplete.
- **Coverage.** Four topics of the guide were not covered (see Table 4), and only two of three specialists answered the follow-up questionnaire.

References

[1] D. Gray, S. Brown and J. Macanufo, *Gamestorming: A Playbook for Innovators, Rulebreakers, and Changemakers*. Sebastopol, CA: O'Reilly Media, 2010.

[2] S. Gibbons, "Empathy mapping: The first step in design thinking," Nielsen Norman Group, Jan. 2018. [Online]. Available: https://www.nngroup.com/articles/empathy-mapping/

A Generative AI Use Disclosure

This appendix describes how generative AI tools assisted in preparing and reviewing this document. The decisions in this report came from the specialist interview, which Moustafa and I conducted, and from our meetings with our supervisor and teaching assistant, including the choice of an empathy map as the analysis method. AI tools helped carry out and express that work; they did not make decisions for us. The author remains responsible for the content of the final document.

A.1 AI Tools Used

- **Anthropic's Claude Opus 5.5**, accessed through Claude Code at low effort.

A.2 How and Where Used

The uses below materially contributed to this document.

- **Organizing the interview data.** AI helped sort through the interview transcript and locate the specific statements, with their timestamps, behind each empathy map entry, each question in Table 3 and each theme. This affected Sections 4 and 6.
- **Drafting and wording.** AI turned my notes and decisions into draft wording for each section, and helped improve sentence structure, clarity and flow. I then directed changes, such as laying the empathy map out with one row per theme, explaining why we built one map for all three specialists, and adding the supervisor's advice in Section 6.3.
- **Formatting.** AI helped with the LaTeX formatting, including the tables and the links into the transcript.
- **Repository management.** Through Claude Code, AI helped prepare the branch and commit for this document.

A.3 Verification of AI-Generated Content

Every AI-assisted passage was reviewed against my intended meaning and the decisions the team had agreed on before it was included. Any wording that introduced claims the specialists did not make, or commitments the team had not agreed on, was removed or rewritten.

Each empathy map entry and each listed question was checked against the original transcript to confirm that the cited statement and timestamp say what this report claims. The "Thinks" and "Feels" columns only hold beliefs and reactions the specialists put into words. The references were checked against their sources, and the report was checked for details that could identify the specialists.

The author is responsible for the accuracy, completeness, consistency, and appropriateness of this document, regardless of whether material was written by a team member or with the help of an AI system.

B Appendix — Specialist Interview Transcript

This appendix contains the transcript of a 43-minute remote group interview held on October 1, 2026, with three domain specialists. The specialists develop software for children with autism, ADHD and other learning needs, and hold doctorates in human-computer interaction and related engineering fields. The interview was conducted by Rayan Nasrallah and Moustafa Moustafa: a 10-minute presentation of the proposed application, followed by a semi-structured discussion. All three participants consented to the use of the transcript for eliciting requirements (see 41:51–42:48 below).

The transcript is de-identified at the participants' request. They are referred to as P1, P2 and P3. Their names, institutions, locations and other identifying details are replaced with bracketed placeholders. A personal health detail is removed. The automatic transcription was corrected where it misheard words, and filler words, false starts and short acknowledgements ("mm", "yeah") were removed. Each turn keeps the timestamp from the original recording (minutes:seconds), which is how requirement sources in this document cite it (for example, P1, 28:37).

Rayan (0:03)

First, I want to introduce myself and also Moustafa. I'm a student at McMaster University. I'm in my final year and I'm studying software engineering with Co-op. So it's an engineering program where we also take a year or so to do an internship program at a company here in Canada. So I'm on my final academic year, and we reached out to you guys because we're doing a project for our final year. It's called the Capstone. And we'll just introduce the project very briefly, and then we want to hear from you guys mostly and get your expertise on the topics we'll be discussing. I'll let Moustafa introduce himself as well.

Moustafa (0:54)

Yes, very similar to Rayan. Also, my name is Moustafa, 4th year software engineering as well. We're in the exact same program. Both kind of had the exact same story. We lived in [location redacted] our whole lives, or for Rayan, it's half of his life, the second half. And then we came to Canada for university. We've been here for five years now. Yeah, Rayan can start.

Rayan (1:16)

Yeah, I'll start sharing my screen. Can you guys see it?

P1 (1:26)

Yes. So both of you are also like final year students in university. Okay.

Moustafa (1:32)

Yeah, we're both on our 4th academic year, but 5th year with the 16 months we took in the middle for the co-op.

P1 (1:39)

Okay, I see.

Rayan (1:41)

Yeah. So last year, we both went for our internship for 16 months, like between our third year and our 4th academic year. So right now we're on our fifth year of university, but final, final year hopefully. Inshallah.

Rayan (2:04)

So I'll just get started. Today, we're going to introduce Squigglr, the application that we're building for our final year engineering project. And over the next 10 minutes or so, we're going to walk you through what we're planning and what we envision for this project moving forward. Let's start with who this app is for. So Squigglr is meant for a pretty broad group of children. That includes kindergarten and primary school kids who are still building their foundational drawing skills and children with autism, ADHD, or other support needs. And one thing we want to be careful about is that we don't want to assume two children with the same diagnosis need the same kind of support. Every child has their own abilities and their own preferences. Furthermore, around each child, you've got the adults who support them, like

parents, guardians, and caregivers, educators and learning support staff, and school administrators. We want Squigglr to help them too. Before we jump into the concept of our application, I want to go through some of the research that shaped our thinking. First, when we say visual motor integration, what we mean is coordinating what you see with how you move. For example, seeing a square and then guiding your hand to draw it. The first paper is Daly, Kelley, and Krauss. They looked at kindergarteners and found that kids with stronger visual motor integration tended to be much better at copying letters legibly. So basically, these skills are closely tied to things like handwriting. It's a correlation though, so it doesn't prove that practicing one improves the other. The second is Taverna and colleagues. They ran a 10-week program with kindergarten and 1st grade children built around short games like completing paths and tracks. And every activity had easy, medium, and hard levels. After the program, the kids' visual motor integration scores went up. There wasn't a control group, so we can't say for sure that the program caused it, but it's a good example of structured practice with graded difficulty, which is pretty much what we're trying to go for. So putting these together, they support the idea behind structured, graded drawing practice, but they don't show that Squigglr itself will get those results. That still has to be evaluated. The next two papers are about why we're paying attention to motor-related needs in children we're designing for. The 3rd paper compared handwriting between autistic children and children without autism. The autistic group had more trouble specifically with forming letters. So basically getting the shapes of the letters right. And within that group, their motor skills predicted how well they wrote. Then the final paper did a systematic review on children with ADHD. They found that fine and gross motor difficulties come up a lot, but the severity ranges from mild to severe. So for us, the big takeaway is really the variation. Not every autistic child or child with ADHD has motor difficulties. And the ones who do are all different. So instead of building one path per diagnosis, we want guidance and difficulty that can be adjusted. And we want kids to be able to practice as much as they need at their own pace. All right, so let's take the first look at the product. Squigglr has three parts connected to each other. First, there are guided activities. That's where kids practice lines, basic shapes, and simple drawings, and they gradually move from tracing to drawing without any guides. Then there's the creative space, where they can just draw freely and use those skills they've built outside of the structured exercises. And lastly, there's practice history, which keeps a record of attempts so parents and educators can actually see what's being practiced. And across all three, you've got adjustable support, history that's saved between sessions, and everything runs in the browser. Now, Moustafa will go through each of these, starting with how an activity progresses.

Moustafa (6:03)

Yes, thank you, Rayan. So we're going to be using the activity of learning how to draw a square as our example here. First, the child would see the demonstration, so where to start and which way to actually go to draw it. Then they trace over a visible outline. After that, we can start taking guidance away step by step. So maybe they only get the corners or part of the outline. And finally, they would draw the square completely on their own with no guides at all. Kids can repeat any of these stages as many times as they want, and the support can be turned up or down. Now, one thing we haven't done yet is decide exactly when a child moves from one stage to the next. So we're not setting any accuracy thresholds or a required number of attempts. We think that's something that should be informed by specialists, not just [inaudible]. So on the next slide, this is a concept of a tracing activity in progress. So in this example, you can see the child has traced three sides of the square, but they missed the right side, for example. Instead of failing the attempt or just finishing it for them, the app highlights the part that's unfinished and gives a short prompt so the child can go back and complete it in the same attempt. There are two main principles here. First, we keep the child's actual drawing. We never swap in a perfect square and count that as them doing it on their own. And the support can be adjusted too. For example, how much of the outline is shown, or if it's just the corners, as mentioned on the slide before. Second, the feedback is based on predefined observations that we are manually deciding. We're not proposing an AI model that gives personalized advice on each drawing. It would just be basic math and algorithms to figure out what's wrong with the shape they drew and comment on it if needed. However, the exact implementation details are still a work in progress. Now, so alongside the lessons, kids also get a free creative space where they can draw whatever they want. In this example on the left, a child has drawn something that looks like a bird. The idea is that the app would take that and tidy

it up just a little bit. So basically the lines would get a bit straighter, the curves get a bit rounder, but it's still the child's drawing. We're not swapping it out for a perfect clip art bird. And if the model can tell what it is, like here it's clearly a bird, it can use that to guide the cleanup. But even if it can't, it will still just clean up the lines and curves a little bit. The way we're planning to do it is we send the drawing coordinates to a lightweight AI model, and then it sends it back slightly adjusted, like it sends back the same thing with slightly adjusted coordinates at the same scale, and we rebuild the drawing inside the application itself. That way, everything stays editable so the child can still move things, resize them, and change the colour or erase them, and they can always restore their original drawing, so they stay in control of their own work. This only happens in the creative space, though. In the structured activities that we talked about in the previous slides, we always keep the child's real drawing. The main goal for this creative drawing area is for it to be fun and to give kids a chance to use their skills that they learned outside of the exercises. Now here is the concept of what a parent or educator might see on our app. For every structured attempt or exercise, the record includes the activity and its guidance level, the child's drawing, any difficulties they faced, and the final outcome. So for example, in attempt one, the child missed 2 sections at first. But when they went back and completed them, the record keeps both of those. Attempt 2 was the same outline guidance, nothing was missed, and then a later attempt, so attempt 3 near the bottom, it said that was completed with fewer guides and it was fine. Adults can look at which activities were practiced, attempts over time, what guidance was used, and the difficulties that kept coming up or got resolved, and the actual drawing of the child. And there are some things we're deliberately leaving out. So no scores, no comparisons with the general population, no diagnostic labels, and no recommendations for special classes or treatment. We also don't want to suggest that fewer attempts automatically means better visual motor integration. The whole point is to just make practice easier to understand and to support better conversations about what a child needs. So now moving on to how we're thinking of building it, Squigglr is planned to be a web application. Basically, a child practices and draws on a compatible device that supports drawing, like an iPad or a tablet. Their attempts get saved to an account, and then adults can review that history later from any device. Part of the reason we went with the browser is that installing separate apps on school-managed devices can be pretty difficult, according to our supervisor who has some experience working with schools in Canada. We'll be targeting supported browsers and compatible devices, but we're not promising offline access or that it will work on every device just yet. Okay, so just to sum up the core idea, we have structured practice with lines, shapes, and simple drawings, progression from tracing to drawing without guides, guidance and difficulty suited to each child, free drawing with automatic refinement, accurate records of attempts and difficulties, and a clear practice history for parents and educators. There are also two extra features that are not part of the main requirements, but we wanted to get your feedback on them as well. The first is camera-based finger tracing, where the shapes and guides show up over a live camera view, and the child traces them with their finger in the air. So instead of having to draw on the screen, they would just do it like this in the air, and it would work the exact same way. The second is taking the same approach of the guided practice we talked about for the shapes and the drawing, but applying it to letters as well. So our main vision is a welcoming place for children to practice drawing at their own pace, with a clear picture of that practice for adults who support them. So that's the whole app idea in summary. Thank you so much for your time. And before we get into our questions for you, please let us know if you have any questions or if anything we chose so far wasn't clear.

P2 (12:07)
Thank you very much.

P1 (12:07)
So thank you very much. Actually, I have a couple of questions. It's a very nice application. One of my questions, actually I have three questions. The first one would be, did you guys co-design this app with educators or therapists or people, like basically the end users who would ultimately utilize this application?

Rayan (12:27)
Moustafa, should I answer?

Moustafa (12:29)
Yeah, go, yeah, for sure.

Rayan (12:31)
So basically, like, we just started this project. We haven't built anything. We haven't even come up with the requirements. So our first kind of thing that we wanted to do is reach out to specialists like you guys that have expertise building apps like this. So the whole point of the engineering project is like we actually build a detailed requirements document over the next month or so. And then we start with the implementation towards December and hopefully finish it towards February and then do a usability test before the end of the second semester, before the end of April. So we're still building the application, like we're still getting the requirements and everything. And the problem with working with kids, I'm sure you guys know, is there's like ethics involved in everything. So our strategy for eliciting requirements was more to go and ask people like yourselves and also like parents and caregivers and so on. So we haven't worked with kids or anything or educators yet, but like this is our first kind of step, if that makes sense.

P1 (12:42)
Okay.

Moustafa (13:41)
Yeah, and also just to add a small note, we also have our supervising professor for the Capstone Project, who also has some experience in this field. So he's partnered with a couple of organizations in Canada, and he has extensive experience with working with kids with autism and stuff like that. So he's going to be able to help us out as well.

P1 (13:42)
Okay. Okay, sounds great. I don't know why I thought that maybe this app was already being developed. The second question I have is a question that you actually proposed in the last slide about the camera, whether they should trace it using the camera. I mean, the camera-based finger tracing, exactly. This idea is nice, but I would say from my viewpoint, I think it might target a very different task. So whenever they're doing it in the air, that might be more of like a gross motor skill. And then whenever they're drawing on an iPad, that's more on the fine motor skills. So this is something that you could probably check with, you know, either therapists or educators and things. But this is my feedback on that. And you could integrate both. I mean, the camera-based finger tracing is more kind of like an augmented reality component that you would be including. And it's very interesting, especially with children who are more neurodiverse. But you just want to focus; they might be targeting different aspects.

Moustafa (14:55)
Okay. Okay, makes sense. But do you think it might be a good idea in general though to have it? Like, do you think there's actually something helpful about this feature if we do end up implementing it?

P1 (15:12)
I do, because a lot of times, even just looking at child development, before they start the very fine tracing, they usually have big markers, big things, and they're kind of tracing it very largely. So this would replicate that. And that kind of works more on the visual motor, even if it's not fine motor, it's more on the gross motor skills. And it helps them also in the visualization. So if they can visualize tracing a square, it might make it easier for them to, you know, trace it when it comes on an iPad and utilize their fine motor skills. So I think it's definitely an important feature to incorporate.

Rayan (15:48)
For now, though, because we might have a limited scope, like we have a certain amount of time to implement this app, do you think our focus with the shapes and the tracing and teaching them to draw is a good starting point and that we should focus on that before implementing this camera feature? Or do you think it should be part of this application from the get-go?

P1 (16:13)
I mean, if you're limited in time, I would say just focus on the first part, and then you could progress later on to extend it to the camera-based finger tracing.

Rayan (16:23)
Sounds good. Thank you for the feedback.

P1 (16:28)
I have a couple more, like some feedback, but I'll give maybe [P2] an opportunity if she has any questions and then I can continue.

P2 (16:40)
Thank you. It's really very nice.

P1 (16:43)
Your voice is very far, [P2].

P2 (16:45)
Can you hear me now? Can you hear me well? So it's really very nice work, both. It's just very inspiring what you're doing. So I agree with [P1] in terms of focusing and leaving the camera and the tracking for maybe something to extend beyond this project. That's one thing. A second aspect, actually, while you were introducing it, I was thinking about my own self when I was very young in elementary school. I've always had difficulty in drawing a line. So when I saw this project, it's very nice. I think it's a very relevant problem. That's first. And my question to you: are you targeting specific groups of children with learning needs, or is it kind of broader? I would like to hear from you, who is your sample population?

Rayan (17:45)
So the idea started for kids with special needs. So kids with autism and ADHD, because the whole thing is each case differs. But from what I've seen personally, even as kids tend to get older, they still struggle with these basic skills that really young kids like toddlers and kids in kindergarten learn to develop early on in their lives, right? So our idea is we're mainly focusing on kids with autism and ADHD, but we also want to extend it to very young children, right? Like this is a skill that, like you said, you remembered when you were young. A lot of kids don't practice this when they're young. It's not integrated into learning. And I feel like it is a very necessary skill. So not just kids with special needs that are a little bit older, but also very young kids as well. We think they can use this application. And one big component of our design, which we wanted to ask about later, was how we can let the guidance and the support be controlled by the caregiver or the parent that's there supporting the activity. Or even if we should have a parent or a caregiver supporting the activity in the first place. So that's where it goes on to.

P2 (19:05)
Thinking about it from an inclusive, universal design perspective, this could be relevant to, yes, I agree, autism and ADHD, but then also children with gross motor and fine motor challenges, and also kids with sight challenges. [Personal account of a childhood vision condition removed.] So I believe this could be a training tool for any child in need of it, regardless of their situation, because there are many conditions that could affect our visual perception, and also affect even our handwriting. So just like when I saw this, I felt like...

Rayan (20:13)
Yeah. That's the plan, hopefully.

Moustafa (20:14)
So similar to what Rayan said, you don't think we should only limit it to children with special needs or autism or ADHD or anything like that? It should be general.

P2 (20:14)
Yeah, because...

P1 (20:27)
Yeah, I agree. I don't think you should limit it, because the skills that you're trying to teach now are applicable to all children, right? But I guess depending on their disability, they might achieve it at different stages in life. But if you go to nurseries that are looking at children at two and three years old, this is what they're focused on.

P2 (20:47)
Exactly. That's true. Actually, there is a saying in the world of accessibility: if you design for people with disability, you are actually designing for all. So whatever serves a certain population could be of use for many other populations. Then, thinking even of elderly people, people who get a stroke at some point in life, this is also very effective. I would see it in a rehabilitation centre, for example.

Moustafa (21:18)
Yeah, I didn't think of it, yeah, that's true.

P2 (21:25)
So the use cases could be endless, really. And I think that's also, if it's a startup, a selling point as well, because it could be put in a rehabilitation centre, it could be put in a kindergarten, it could be put in a school or at home.

Rayan (21:45)
Inshallah.

Moustafa (21:47)
Something happened to your mic. Yeah, yeah. I think it disconnected.

P2 (21:57)
We can see you, but we can't hear you.

Rayan (22:00)
Hello!

Moustafa (22:03)
Talk again.

Rayan (22:04)
Okay. Can you guys hear me?

Moustafa (22:06)
Yeah, that was it.

Rayan (22:07)
Yeah, yeah, I forgot what I was saying. Yeah, we wanted to ask actually if at some point we are able to build this app, would you guys, because you guys actually do research, would you guys see yourselves kind of looking into this area, or using this app, or doing further research? Like, if we just start with the idea, like a proof of concept, this app could serve as a proof of concept. Would you guys be able to take it beyond that in the future? And hopefully maybe in February, be able to actually get children to use it and do some observation, not necessarily a trial or research, but just observation activities, because it's difficult for us to do it here in Canada. That's why I wanted to ask.

P2 (22:56)
Yeah, but actually, it's equally difficult [here]. You need to get the ethical approval and the process is quite... Yeah, I think the laws have been tightening over the last 10 years, I would say, especially in this part of the world and in research. But it's easy for us to obtain approvals because we have experience, and an approval would take about three to four months really to obtain. But then actually, as you also speak, I am aware of a group at [another local university] that is looking at handwriting and dyslexia and other learning [needs]. So there is also a very active group there that looks at learning needs and technology, the use of image processing in particular. I can share their contact with you if you're also interested to meet more expertise in the field. And I also wanted to say something else. Well, I forgot, but these are the two points from my side, but this is really nice, very nice work.

Moustafa (22:59)
Yeah, I think it's difficult everywhere.

Rayan (24:03)
Thank you.

Moustafa (24:03)
Thank you so much.

P1 (24:03)
Yeah, absolutely. I think it's great work. I think it's much needed. And I mean, we would be happy to kind of help guide you. I think there is some merit in trying to, you know, pilot or test this application. Of course, you would need an IRB, but maybe you could do this where you're at, through your university, right? And we could support you through the process and kind of help guide you. So I think it's a great application.

Rayan (24:26)
Yeah, that would be perfect. Yeah. Thank you guys. Appreciate it.

P2 (24:31)
Thank you.

Rayan (24:32)
[P3], you.

P3 (24:34)
Yes, I just want to ask one thing. Honestly, I'm curious, I'm not sure if I missed it or not while you were presenting. Will you ask the children to hold the pen in order to do the line, like the drawing, or will it be their finger? Like, will it be touch, or are you thinking of a pen?

Rayan (24:56)
I think it can be either. We haven't really got into that decision. Our initial thought was a pen. Like, I have this iPad here. So like this, for example. And then maybe as well, there's attachments you can buy for these that make the grip wider. Because I know some kids have trouble holding it like this. So I know there's grips that you can buy that make it large. So it's very flexible. I feel like it should support tracing with the finger and pen. But what do you think?

Moustafa (25:03)
Thinking about it now, a pen is probably not a good idea, right?

P3 (25:11)
Exactly, yeah.

P3 (25:36)
And that's why I ask, because as [P1] was saying and [P2] also was saying, regarding the fine motor skills of the children, like how they will hold the pen? Will they hold it from the top, or some will hold it like this? You know, I don't have a pen near me, but they will hold it like this. So this might affect how they will draw, which might affect maybe the usability of the app itself. So if your goal is not about the fine motor or something, we could remove the pen. If your goal is more about the drawing itself, only the drawing idea, I guess you can use the finger thing. Not sure if [P2] and [P1] agree with me on this idea, but just to make your app more reliable in real life for the children.

P1 (26:35)
So you mean to draw with their finger on the application?

P3 (26:40)
Yeah, it will be like a touch. They will draw as Moustafa was saying in space using the camera. So it will be the same but on the tablet or iPad itself, rather than holding a pen and having more challenges.

P1 (26:46)
In the air, yeah. That may be a start. I see that. That's a good option for a start. But again, if the idea is to kind of improve the fine motor skills, they have to practice with being able to hold something. So

usually they start with something large, because it's easier for their hands, and then they have to go to the smaller ones, like pencils. That's more difficult, so that actually takes some training for them to do. So you could embed difficulty levels, you know, where they start with their finger, which is the easiest, and then with a larger kind of marker or pencil, and then something smaller that's more reflective of what they would use in schools.

Rayan (27:31)

So on that point, we had a question about how we should adjust, because in our thought process, we don't want the app to judge when a child is ready to move from one stage to the next. So the example we gave throughout the slides was the square. But of course, a square is a very simple shape. And then they could move to triangles and then stars and more complex objects that they can trace and then draw by themselves, right? And we'd have an accuracy threshold. But our concern was, should we dictate when the child is ready to move from a square to a triangle or a square to a star? Or is that something we should leave fully to the supervising adult during the activity? And what determines when a child is ready to move on? Because now you introduce an extra degree of complexity, which is holding an object in and of itself; decreasing the size of that is also progression, but that's not something we can necessarily track.

P1 (28:37)

So if you're going to look at that, then you're going to open up a door of assessments, right? Because there needs to be assessment. And I think that, given your time frame, it might be too complicated at this stage. I would keep it user interactive, so the user can decide whether they're going to go from, I don't know, level one, level 2, level 3, right? And then ultimately who decides what they're going to do would be either their parent or the educator or somebody who knows where their level is at. And then they could tell them, you know, you need to practice level one until you're proficient in it, and then they can ask them to move. But if you want the app to do it automatically, that's going to be more advanced.

P2 (29:18)

Actually, just something we had in one of our projects: we had an app for children to learn vocabulary, and we co-designed it with a centre [location redacted] with speech therapists in the centre, and we allowed the teacher to kind of personalize the journey for each child separately, and that's not really automated. The teacher would have a back-end kind of interface to do that. But as [P1] said, that actually will bring you to the other side of the story, which is assessment. But then you might want to think about just allowing the teacher or the caregiver to kind of change the activity and giving them a back-end view of the progress. But then, thinking also, if you do that, then one would ask, oh, what kind of assessment are you applying here?

Rayan (30:20)

Yeah. So I think maybe the best thing is to do what you guys recommend, which is to kind of leave it to the caregiver or the supervising adult, and we shouldn't worry about assessment right now. It shouldn't be in the scope given the time frame. Maybe something in the future, Moustafa, I don't know if you agree.

Moustafa (30:38)

Yeah, I feel like it's very good to have, but maybe not in these short months. It would be nice, but I don't think we have time to do it right now. So right now, as you guys recommended, it would probably be the adults who would say, okay, this is fine for now, move on to level 2 or level 3, instead of us having to figure out how to map when this child is ready for level 2 or level 3. We can figure that out later.

Rayan (30:52)

Hundred percent, yeah.

Rayan (31:10)

Speaking of that, in the history that we showed, what statistics or what values or what information would actually be useful to the parent or caregiver, so they can say, okay, this progress history is actually benefiting my judgment of whether the child is ready to progress? For example, should we have a

percentage, like, okay, the accuracy threshold is above 80%, we passed it, but we're still showing you that this is the accuracy of the kid's drawing?

P2 (31:42)
That's a very good question. Really, I think this is a question to ask the therapists or the teachers, if there are certain defined milestones in the literature that you could kind of have as your benchmark or reference there. And that's something to discuss really with, I think, either an OT, an occupational therapist, or an educational specialist.

Rayan (32:12)
Yeah. Do you guys know any occupational therapists or educational specialists that we can speak to about something like that, that you could connect us with?

P2 (32:21)
I think the easier way to do that is reaching out to private centres [location redacted]. I think [P3] can share them with you step by step. They are very good, they are very approachable. You might want to introduce this, and they are very interested in technologies. So, maybe that.

P1 (32:42)
And [a local hospital] as well. They have a therapy program where they have occupational therapists as well.

Rayan (32:50)
We're scared that if we email them, we might scare them or something. They might start talking about ethics and stuff, but it's just a small project. It doesn't seem like it, I guess, when we reach out to them, but it's hard to kind of...

P1 (33:00)
Well, the ethics only comes in whenever you say that you want to test your product, you want to meet with the kids and stuff. But if you're asking just to meet with them and get their expertise and get their feedback, it should be okay as long as you're not really interacting with the children at this point. Just to get feedback through a meeting, like what we're doing now, it should be fine.

Rayan (33:26)
Okay. Sounds good. Moustafa, do you have any more questions you wanted to ask?

Moustafa (33:34)
Yeah, this was on a point that we were mentioning before. So with deciding when the child goes on to the next level and stuff like that, a similar question: how should this app be made? Should this app be built in a way that the child by himself can go in and kind of know how to play around and move from level one to level 2, or even go to the creative mode and draw? Or are we expecting that there's always a parent present? Because we've never worked with kids with disabilities, or not even kids with disabilities, just children in general, very young people. We don't know how to give feedback, like, oh, good job, now try this. Or how do we tell them to draw a square? Or how do we tell them to do this? Or are we assuming that there's a, yeah?

P1 (34:13)
That's a very good question. Yeah, so to answer the first part of your question, I think you should design the application so that it's child-friendly. Because if you're going to design it in a way that the parent has to keep, you know, interfering, parents are going to quickly get annoyed, and, you know, if they give them the iPad to practice, they want them to practice without their constant interference. So then you have to think about how you design the application so that it's child-friendly, meaning that, you know, both the child and the adult can understand the interaction. So a child who's three years old maybe doesn't know how to read yet, but, you know, kids these days are very tech-savvy, so they know how to navigate applications and things. But if you look at the games that they play, it's usually not very text heavy or oriented, so they have images and icons that they navigate in order to go and play. So you would have to design it from the get-go in that kind of manner. Adults obviously will know how to

navigate through and install and do the initial parts, but you know, you should also design it in a way so that children can navigate and be able to enter the lesson that they're on, be able to practice, to restart, to go back and forth. Because otherwise, if you make it too high level, where they require constant adult supervision, I think you're going to limit the usability of it.

Moustafa (35:31)
Okay, so it should be somewhat expected that the child can use it alone, play around with it.

P1 (35:36)
Yes, and then, yes.

P2 (35:36)
I think also, sorry, [P1]. It's just, I think also it's your decision on what you want the app to look like. Like, you know, there are many drawing apps in the App Store, right? For children, for mainstream, or maybe not even in mainstream school, it could be used in a centre. But it's your decision to say whether you want this app to kind of follow a process and have, yes, a child as a user, but then the caregiver can give them the access. So I think these different workflows, you can put whatever you like, because there is no right and wrong, sort of. There's only the way you present it and the angle in which you're presenting it.

Moustafa (36:22)
I see, I see.

P1 (36:26)
No, no, there's no problem at all. I think the next step would be for you guys to meet the end users and the stakeholders, which would be the educators and occupational therapists. They can also give you more feedback. And they're the ones who will be able to guide you, because we're developing similar, not similar, but I mean, similar types of applications, one for occupational therapy. And even the way that we give them positive reinforcement, like if they get something correct, what we say and how we say it and the colours that we use. I mean, we go through the literature, of course, but we also get the feedback from, you know, the experts, because we're doing the design and the technology aspect, but they're the ones who deal with the patient side or the children's side on a day-to-day basis. So these kinds of design aspects, you can build bit by bit.

Moustafa (37:15)
I see. Okay, thank you so much. Rayan, you had something to add.

Rayan (37:18)
I just had a follow-up, yeah, so... The first thing is, what do you guys think about voice, like dictation, like voice support for the kids? Like saying, oh, you did a good job, with like some sound? Or do you guys generally see that that's not a good idea?

P2 (37:39)
No, actually, it is a very good idea. We had an app, the vocabulary learning one, that had a voice input as well. And multi-modality in general, like having two modalities to communicate, and then the user could choose; he could turn off the audio. So multi-modality is very much encouraged in the field of design, right? If you want to design an app... now if you look at the Apple products and the phones in particular, they also have the haptic modality, so the phone can shake sometimes if there is a notification. So multi-modality is always encouraged, even in the realm of inclusive design and universal design. It's always encouraged, but it's just, as [P1] said, designing it carefully, getting the feedback from the therapists, and even, if you have the chance, getting the feedback from the users themselves. Even if you cannot reach the children as a sample or a group, you could reach an adult group with ADHD and autism. And if you want to expand it to different adult groups, then that's also useful, because then you based your decision on real experiences.

Moustafa (38:58)
It's a very good idea, okay.

Rayan (38:58)
Okay. And then I have a follow-up to that as well. In general, what design choices in apps tend to cause problems for kids with autism and ADHD? And also, what are best practices for these types of apps? Because you all have worked on HCI and these types of apps, making sure the interface is specific to their needs. So what general patterns do you guys recommend we should follow?

P2 (39:26)
So yeah, this is a very good question. I teach an elective course for the undergrads in the spring on ICT accessibility, so how to design accessible interfaces and apps, and the main set of guidelines is called the Web Content Accessibility Guidelines. We call it the WCAG. And it's very famous. It actually is adopted as law in many countries, including the US, and [location redacted] adopted it last year. So I think start with the WCAG and then move forward to more specialized guidelines too. Start with the WCAG and then look at design guidelines for autism in particular; we call them heuristics. So heuristics for autism, app design for autism. There is a lot in the literature; just write in Google Scholar "heuristics for app design for autism" and you'll find a lot. But I would suggest starting with the WCAG and then moving to the more detailed ones.

Rayan (40:39)
Okay, perfect. Thank you. Thank you.

Moustafa (40:39)
Your course would be very helpful for us right now, the accessibility course. I wish, I wish.

P2 (40:44)
Yeah, you're welcome to attend, welcome to come [location redacted].

Rayan (40:44)
Yeah. Yeah. Hopefully, that's the dream. Inshallah. Do you guys have any more comments about that? Like, [P1], [P3]? Do you guys have any advice or any patterns that you guys think of?

P1 (41:08)
I mean, from my end, no. I think what you're doing is really great and it's very timely, and we would be happy, maybe after a couple of weeks or a month, to meet again. And if you have questions, we can again give you our view and feedback, just to kind of help guide you through the process.

Rayan (41:26)
Yeah, that would be perfect. I mean, we have a lot more questions, but because we're running short on time, would it be okay if we sent you guys a form or something that you can answer on your own time, whenever you have time?

Moustafa (41:26)
That would be amazing, yeah. It would probably be just very, very short questions.

Rayan (41:44)
Like, they're very specific.

P2 (41:49)
Yes, of course.

P1 (41:51)
Yeah, that's fine.

Rayan (41:51)
Okay, perfect. Well, we'll create a form then, and we'll send that out to you guys, and it has some more questions that we didn't really get to ask. And also, the whole purpose of this interview, like we said, was to kind of get your feedback and then elicit requirements from that. I guess now I'm realizing we have a lot more work to do. We have to talk to some more people before we can actually elicit the requirements properly. But I was going to ask you guys if you could sign a form, a consent form, that we can use this

transcript to basically do that work. We'll send it to you guys as well, with the questions, at the same time, by email, if that's fine.

P2 (42:37)
Yeah, of course.

P1 (42:38)
No, it's fine.

Moustafa (42:39)
Yeah, it's basically just saying that we can use the transcript of what we talked about here to elicit the requirements, and that we had this meeting, basically. That's all.

P3 (42:40)
Sure, no problem.

P1 (42:47)
Sure.

Rayan (42:48)
Okay, perfect. Thank you guys so much.

Moustafa (42:49)
Thank you so much. Also, does anyone have any other questions before we end?

P1 (42:52)
No problem.

P3 (42:56)
Thank you so much. That's amazing work.

P1 (42:58)
No problem. Best of luck [inaudible].

Rayan (43:00)
Well, we'll be reaching out to you guys again, hopefully in the near future. Inshallah.

Moustafa (43:00)
Thank you guys so much. This was very helpful. Inshallah.

P3 (43:06)
Inshallah, Inshallah.

Moustafa (43:08)
Thank you so much. Thank you so much.

P1 (43:09)
Thank you.

Rayan (43:10)
Thank you guys for your time. Bye-bye.

P3 (43:11)
Thank you, bye.

P1 (43:14)
Bye.